Let Skills Evolve Collectively with Agentic Evolver

SkillClaw

Collective Skill Evolution for Multi-User LLM Agent Ecosystems

arxiv.org/abs/2604.08377

Agent Skills Remain Static After Deployment

Current Paradigm: Static Skills

• Skills manually installed from hub

• Never updated from runtime experience

• Failures rediscovered independently

• No system-level knowledge accumulation

Each user re-invents solutions

SkillClaw: Evolving Skills

• Continuous runtime learning

• Cross-user trajectory aggregation

• Shared evidence base

• Automatic skill evolution

Collective intelligence

The Problem in Practice

Similar tasks recur across different users over time

Same patterns of failure and recovery are repeatedly observed

Memory-based methods remain tied to specific instances

Skill libraries treated as static resources

Closed-Loop Pipeline: From Multi-User Interaction to Shared Skill Evolution

Key: Agentic evolver analyzes skill-specific groups through three open-ended reasoning stages

Evidence → Attribution → Evolution → Distribution

Agentic Evolver: Evidence → Attribution → Evolution

1. Evidence

Analyze trajectories to identify behavioral patterns across multi-user sessions

Input: Structured session trajectories preserving full action–feedback causal chains

Pattern detection from diverse user contexts

2. Attribution

Diagnose root causes of failures and successes to understand why skills work or break

Identifies: Generalizable improvements vs. idiosyncratic fixes

Single user rarely generates enough signal alone

3. Evolution

Propose skill mutations — refine existing or create new skills

Open-ended agent reasoning over evidence

NOT predefined rules

KEY: Agentic evolver drives skill updates through autonomous reasoning, not rule-based logic

Cross-User Trajectory Aggregation Exposes Skill Behavioral Boundaries

Compatible Claw-Style Systems

- OpenClaw
- CoPaw
- IronClaw
- PicoClaw
- ZeroClaw
- NanoClaw
- NemoClaw

All exercise the same skills under diverse contexts

Cross-User Aggregation

User A
Context 1

User B
Context 2

User C
Context 3

Trajectories Aggregated

Shared Evidence Base

Complementary Signals

Success patterns + Failure modes across diverse contexts

Critical Insight:

A single user rarely generates enough signal to separate a generalizable improvement from an idiosyncratic fix.

Case Study: Multimodal Creative Pipeline Skill — 6 Days of Evolution Attempts

1 of 6 Candidates Accepted — Conservative Verification Process

Day	Candidate Skill	Validator	Next-Day Action
1	validate-tmp-workspace-inputs	Accept	Upgrade to new best pool
2	multimodal-input-validation-and-setup	Reject	Keep current best pool
3	multimodal-creative-task-pipeline	Reject	Keep current best pool
4	multimodal-creative-task-pipeline (impr.)	Reject	Keep current best pool
5	multimodal-creative-task-pipeline (impr.) + validate-required-input-files	Reject	Keep current best pool
6	multimodal-creative-task-pipeline (cand.)	Reject	Not admitted to next cycle

Accepted Rejected

Takeaway: Only 1 acceptance over 6 days demonstrates strict regression prevention

Case Study: Git Push Auth-Fallback Skill — Iterative Refinement Over 6 Days

3 of 6 Candidates Accepted — Successful Iterative Refinement

Day	Candidate Skill	Validator	Next-Day Action
1	git-push-with-auth-fallback	Accept	Add to Safety best pool
2	git-push-with-auth-fallback	Accept	Keep updated best pool
3	git-push-with-auth-fallback + git-clone-to-directory	Accept	Keep current best pool
4	(none)	Accept	Continue same best pool
5	git-push-with-auth-fallback	Reject	Keep current best pool
6	git-push-with-auth-fallback	Reject	Not admitted to next cycle

Accepted Rejected

Takeaway: 3 successful improvements before plateau — evidence supports iterative refinement when warranted

Key Takeaways: Three Properties That Distinguish SkillClaw

1. Collective Evolution

Knowledge from individual interactions contributes to a shared, continuously improving skill ecosystem

Multi-user trajectories → shared improvement

2. Fully Automatic

Skill evolution driven by runtime interaction without manual curation or explicit user intervention

No human in the loop

3. Agentic Evolution

Skill updates produced through open-ended reasoning rather than predefined update rules

Autonomous agent reasoning

Implications for Building Continuously Improving Agent Systems

SkillClaw compatible with OpenClaw, CoPaw, IronClaw, PicoClaw, ZeroClaw, NanoClaw, NemoClaw

Evaluated on WildClawBench using Qwen3-Max — substantial improvements across tasks